

A

B

C

D

PMP® FORMULAS CHEAT SHEET

1. EARNED VALUE MANAGEMENT (EVM)

PV (Planned Value)	Budgeted cost of work scheduled
EV (Earned Value)	Budgeted cost of work performed
AC (Actual Cost)	Actual cost of work performed
CV (Cost Variance)	$CV = EV - AC$
SV (Schedule Variance)	$SV = EV - PV$
CPI (Cost Performance Index)	$CPI = \frac{EV}{AC}$
SPI (Schedule Performance Index)	$SPI = \frac{EV}{PV}$
EAC (Estimate at Completion)	
a) $EAC = BAC / CPI$	
b) $EAC = AC + (BAC - EV) / CPI$	
ETC (Estimate to Complete)	
a) $ETC = EAC - AC$	
b) $ETC = (BAC - EV) / CPI$	
VAC (Variance at Completion)	$VAC = BAC - EAC$
TCPI (To Complete Performance Index)	
a) $TCPI = (BAC - EV) / (BAC - AC)$	
b) $TCPI = (BAC - EV) / ETC$	
BAC (Budget at Completion)	$BAC = \text{Total budgeted cost}$

2. SCHEDULE MANAGEMENT

Forward Pass	
ES (Early Start)	$ES_i = \max(EF \text{ of all predecessors})$
EF (Early Finish)	$EF_i = ES_i + \text{Duration}_i$
Backward Pass	
LF (Late Finish)	$LF_i = \min(LS \text{ of all successors})$
LS (Late Start)	$LS_i = LF_i - \text{Duration}_i$
Float Calculations	
Total Float (TF)	$TF_i = LS_i - ES_i$ $= LF_i - EF_i$
Free Float (FF)	$FF_i = \min(ES \text{ of all successors}) - EF_i$
Independent Float (IF)	$IF_i = \max(0, \min(ES \text{ of all successors}) - \max(EF \text{ of all predecessors}) - \text{Duration}_i)$
Critical Path	
Path with zero total float	
Project Duration = $\max(EF)$ of all activities	
Note: $i = \text{activity}$	

3. COST & RESOURCE MANAGEMENT

Cost Variance	
$CV = EV - AC$	
Cost Performance Index	
$CPI = \frac{EV}{AC}$	
Budget Variance	
$BV = BAC - EV$	
Estimate at Completion	
$EAC = AC + \frac{(BAC - EV)}{CPI}$ or $EAC = \frac{BAC}{CPI}$	
Estimate to Complete	
$ETC = EAC - AC$	
To Complete Performance Index	
$TCPI = \frac{(BAC - EV)}{(BAC - AC)}$ or $TCPI = \frac{(BAC - EV)}{ETC}$	
Resource Utilization	
$\text{Resource Utilization (\%)} = \left(\frac{\text{Actual Units}}{\text{Available Units}} \right) \times 100$	

4. QUALITY MANAGEMENT

Defect Density (DD)	
$DD = \frac{\text{Number of Defects}}{\text{Size of Product}}$	
Defects Per Million Opportunities (DPMO)	
$DPMO = \left(\frac{\text{Number of Defects}}{\text{Opportunities}} \right) \times 1,000,000$	
Yield Percentage	
$\text{Yield (\%)} = \left(\frac{\text{Opportunities} - \text{Defects}}{\text{Opportunities}} \right) \times 100$	
Cost of Quality (CoQ)	
$\text{CoQ} = \text{Prevention Cost} + \text{Appraisal Cost} + \text{Failure Cost}$	
$\text{Failure Cost} = \text{Internal Failure Cost} + \text{External Failure Cost}$	

5. RISK MANAGEMENT

Risk Exposure (RE)	
$RE = \text{Probability of Occurrence} \times \text{Impact}$	
Expected Monetary Value (EMV)	
$EMV = \sum (\text{Probability}_i \times \text{Impact}_i)$	
Decision Tree - Expected Value (EV)	
$EV = \sum (\text{Probability of Path} \times \text{Payoff of Path})$	
Monte Carlo Simulation	
Use probability distribution to compute mean, standard deviation, $P(X \leq x)$, $P(X \geq x)$	
Risk Priority Number (RPN)	
$RPN = \text{Probability} \times \text{Impact} \times \text{Detectability}$	

6. PROCUREMENT MANAGEMENT

Total Cost of Ownership (TCO)	
$TCO = \text{Purchase Price} + \text{Operating Cost} + \text{Maintenance Cost} + \text{Disposal Cost}$	
Make or Buy Analysis	
$\text{Net Benefit} = (\text{Cost of Purchase} - \text{Cost to Make})$	
Choose option with higher net benefit	
Seller Performance Index (SPI_p)	
$SPI_p = \sum (\text{Weight}_i \times \text{Rating}_i)$	
Where $\sum \text{Weight}_i = 1.0$	

7. STAKEHOLDER MANAGEMENT

Stakeholder Influence / Power (0-5)	
Stakeholder Interest (0-5)	
Stakeholder Impact Index (SII)	
$SII = \frac{(\text{Influence} + \text{Interest})}{2}$	
Engagement Impact	
$\text{Engagement Impact} = SII \times \text{Engagement Level}$	

8. COMMUNICATION MANAGEMENT

Communication Effectiveness (CE)	
$CE = \frac{\text{Messages Received \& Understood}}{\text{Total Messages Sent}} \times 100$	
Communication Efficiency (C_{eff})	
$C_{eff} = \frac{\text{Useful Communication}}{\text{Total Communication}}$	
Information Overload Index (IOI)	
$IOI = \frac{\text{Volume of Information}}{\text{Capacity to Absorb}}$	

9. BENEFIT MANAGEMENT

Return on Investment (ROI)	
$ROI (\%) = \frac{\text{Net Benefits}}{\text{Total Investment}} \times 100$	
Benefit Cost Ratio (BCR)	
$BCR = \text{Total Benefits} / \text{Total Costs}$	
Payback Period (Years)	
$\text{Payback Period} = \text{Initial Investment} / \text{Annual Net Benefits}$	
Net Present Value (NPV)	
$NPV = \sum_{t=0}^n \frac{(\text{Benefits}_t - \text{Costs}_t)}{(1+r)^t}$	
Where $r = \text{discount rate}$	

10. FORMULA SUMMARY TABLE

Area	Formula	Key Use
EVM	$CPI = EV / AC$	Cost performance
EVM	$SPI = EV / PV$	Schedule performance
EVM	$EAC = BAC / CPI$	Forecast at completion
Schedule	$TF = LS - ES$	Schedule flexibility
Risk	$RE = P \times I$	Risk exposure
Risk	$EMV = \sum (P \times I)$	Risk decision making
Quality	$DPMO = (D / Opp) \times 10^6$	Process quality
Procurement	$TCO = \sum \text{All Costs}$	Total ownership cost
Stakeholder	$SII = (I + In) / 2$	Stakeholder priority
Benefits	$ROI = (B / I) \times 100$	Investment return

PV: Planned Value | EV: Earned Value | AC: Actual Cost | BAC: Budget at Completion
 ES: Early Start | EF: Early Finish | LS: Late Start | LF: Late Finish | TF: Total Float
 P: Probability | I: Impact | D: Defects | Opp: Opportunities | B: Benefits | I: Investment
 Σ: Summation | r: Discount Rate | t: Time Period

11. COMMON DISTRIBUTIONS (RISK)

1. Beta Distribution (PERT)	
$E = \frac{(O + 4M + P)}{6}$	
$\text{Variance} = \frac{((P - O))^2}{6}$	
Where: O = Optimistic M = Most Likely P = Pessimistic	
2. Triangular Distribution	
$E = \frac{(O + M + P)}{3}$	
3. Standard Deviation	
$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n - 1}}$	
Where $\mu = \text{mean}$, $n = \text{number of observations}$	

12. NET PRESENT VALUE (NPV) DETAILS

$NPV = \sum_{t=0}^n \frac{(\text{Cash Flow}_t)}{(1+r)^t}$	
$\text{Cash Flow}_0 = -\text{Initial Investment}$	
$r = \text{Discount Rate (required rate of return)}$	
$NPV > 0 \rightarrow \text{Accept}$	
$NPV = 0 \rightarrow \text{Break Even}$	
$NPV < 0 \rightarrow \text{Reject}$	

rPlan

AT 39, MERMAID 1 CHS LTD., SEC-11,
 CBD-BELAPUR, NAVI MUMBAI - 400611

Legal Owner

QR Code



Draw electronic code

Drawn By:

Drawn Date

Draw Purpose Code :-

Status of Approval :-

Dimensions as Mentioned

Engg Code :-

Backup Info Code :-

Angle of Projection :-

Checked Approved by

Checked Approved Date

Rev

DWG No.

XXXX

SCALE:-

SHEET

Description:

PMP Formulae for Project Control

Preliminary Doc Under prep

Statutory Note:

This document is prepared by rPlan and is the sole proprietary of the company. Its purpose is educational and no references within this document are for Engineering or Design correlation with the applicable codes. For any clarifications beyond the intended purpose viewers can always correspond to Plan via its company website. This drw is not for duplication without company permission!